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Build Practical AI Agents with **h2oGPTe**

SPEAKER

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h2o.ai/university

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Why Agents Matter Today

- 78 % of companies now use AI in at least one business area
- GenAI users report ~5 % productivity gain ≈ 2.2 h/week saved
- Daily GenAI users report higher productivity, compensation, and job stability

Sources: Stanford HAI (2025) • St. Louis Fed (2025) • PwC Global Workforce Survey (2025)

What if you could build your own AI agent in minutes?

Are you ready to:

- Build an AI agent or two?
- See how agents can optimize your everyday work?

Take a look at this live example:

🔗 https://andreea-turcu.github.io/holiday-company-gift-card-agents-demo/index_v2.html

Are you ready to learn how this entire demo app was built in just a couple of minutes from scratch?

AGENDA

1. Introduction
2. Build Your First Agent in 10 Minutes
3. Upgrade Your Agent with Smart Capabilities
4. Certification and Conclusions

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1. **Introduction**
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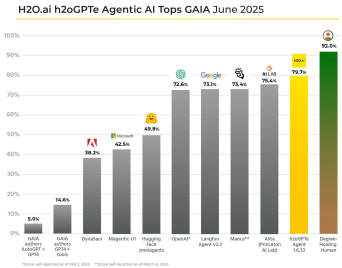
No one is closing the last mile between AI and enterprise value.

- **World's Best** Agentic AI and Deep Research
- **Built-in** Evaluation Toolkit to benchmark and validate agents



- Offers **Sovereign AI** with leading Generative AI, AutoML, Deep Learning for clouds and air gapped deployments

- **Deep Domain Expertise** in BFSI, Telco and Public Sector with **Proven ROI**
- **Ready-to-use** vertical reference designs can be customized for your use case

- **Validated on** NVIDIA Enterprise AI Factory and DELL AI Factory

BLOG ANNOUNCEMENT
WHITEPAPER

* IRAP and FedRAMP High in process

Grandmaster Led Time to Value

H2O.ai embeds the world's top data science talent directly into end customer's projects - H2O.ai's unmatched team of top Kaggle Grandmasters (5% of the world's elite), domain experts, and SMEs, co-creates and develops agents with the customer, giving them a unique edge no other software vendor can offer.

KAGGLE GRANDMASTERS World's Best Data Scientists

H2O.ai KGM HIGHLIGHTS

- Released foundational language model H2O Danube3
- Released foundational vision model H2O Mississippi
- Building autoML for fine-tuning
- Top spots in competitions:
  - Detect AI Text
  - LLM Prompt Recovery
  - Human Preference
  - Predictions
  - PII Detection
  - Arial Data Challenge

1 Triple GM
4 Double GMs

3 World #1

2 Top 10 Globally
5 Kaggle Masters

H2O products are backed by **5% of the World's Data Science Grandmasters** and a **Team of Experts** who are relentless in solving critical problems.

What is an Agent?

An agent is like a smart robot that can:

- Think (using AI)
- Use tools (like a calculator, search engine, or drawing program)
- Do multiple steps by itself to complete your task

Creating an Agent = Giving the Robot Superpowers

Why Agents Matter Today

Challenges teams face

- Repetitive manual tasks
- Slow turnaround for analysis and prototypes
- Fragmented tools and inconsistent outputs

What agents change

- They reason and follow multi-step processes
- They generate structured, reusable outputs
- They integrate with tools and data inside your environment

Why Agents Matter Today

Challenges teams face

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What agents change

- They reason and follow multi-step processes
- They generate structured, reusable outputs
- They integrate with tools and data inside your environment

Why it matters now:

- Agents move AI from assistance to real execution.
- They unlock consistency, speed, and automation that traditional models cannot provide.

We deliver ready-to-use vertical agents that can be customized with your data, tools, and models for faster time to value.

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OUT-OF-THE-BOX AGENTS, READY TO BE CUSTOMIZED

- General AI Assistant
- Deep Research
- Workflow & Task Management
- Web Search
- Tool Builder
- MCP-Only Runner

VERTICAL TASK AGENTS

- Code Agent
- Data Science Agent
- Data Extraction Agent
- Data Assembly Agent
- Data Ingress Agent
- Data Egress Agent
- Data Enrichment Agent
- Data Quality Assurance Agent
- Data Load Agent
- Requirements Analysis Agent
- FinCrime Agent
- Procurement Agent
- Fraud Agent
- Cyber Security Agent

Use your agent as is,
or customize for your exact needs:

- Modify the System Prompt to push for specific tasks and goals
- Enable and disable any of 25+ out-of-the-box tools
- Add your own tools:

Existing MCP servers	Custom MCP servers	Browser tools	Prompting + python tools

- Choose if you want to emphasis speed / low cost or accuracy
- Choose your LLM:

Bring your own external model	Use local models like oss-gpt, llama 4 scout and more

- Enable guardrails, PII detection

Save the settings so that any users that come to this use case get an agent that works just how you want it to

- Role based access control to make sure the right people can use or modify each use case
- Access from the H2O UI or integrate with Rest and Python APIs

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Connect to Enterprise h2oGPTe

Go to the main website: h2o.ai
Type: What is Enterprise h2oGPTe in the Chat bar
Press Enter
You will be directed to the https://h2ogpte.genai.h2o.ai/chats/ for getting the answer
You can Sign in with your Google, Apple or Github account to have access to more capabilities.

Follow along to create a working agent using built-in tools

Enable the Agent

1. [CLICK: + New Chat]
2. [CLICK: Customize button]
3. Open Config tab
4. Scroll to Use Agent
5. [TOGGLE: Use Agent = ON]
6. Watch the UI expand (Agent Type, Accuracy, Tools appear)

Follow along to create a working agent using built-in tools

Configure Your Agent

In the Config tab:

1. Agent Type: General
2. Agent Accuracy: Standard
3. Under Tools, enable only:

• python
• internet

4. Disable all other tools
5. [Close the panel]
6. Confirm “General” badge in chat

Follow along to create a working agent using built-in tools

Test Your Agent

Copy and paste this prompt:

I'm planning a holiday party for 15 people with a $300 budget.
Can you help me create a complete party plan including:
- Budget breakdown (food, drinks, decorations, entertainment)
- 3 easy appetizer recipes
- A shopping list
- A timeline for party prep
Make it festive and fun for a Christmas theme!

Then **[PRESS: Send]**

Follow along to create a working agent using built-in tools

Generate Structured Output

Copy and paste:

Perfect! Now regenerate that party plan as a JSON object with this schema:

```
{
  "party_details": {
    "theme": "string",
    "guest_count": number,
    "total_budget": number
  },
```

Then **[PRESS: Send]**

Follow along to create a working agent using built-in tools

Validate JSON

1. **[COPY the JSON]**
2. Go to **jsonlint.com**
3. **[PASTE and validate]**
4. Look for: “Valid JSON”

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Upgrade Your Agent with Smart Capabilities

To begin:

- Stay in the same chat as before
- **[CLICK: Customize button]**
- **[CLICK: Config tab]**
- Verify **Use Agent = ON**
- Confirm:

– Agent Type: **General**

– Agent Accuracy: **Standard**

– Tools: python, internet

Upgrade Your Agent with Smart Capabilities

In **Config** tab:

1. Scroll to Tools
2. Enable:
   - python
   - internet
   - **Rag Text**
3. Drag and Drop a document (example: “Company Policies”: https://triagelogic.com/wp-content/uploads/2018/06/Company-Policy-and-Procedure-June-1.18-V6.0.pdf)
4. Return to **Config** and check in Collection the document and verify tools

Upgrade Your Agent with Smart Capabilities

In **Prompts** tab:

• Open **System Prompt** and B the current Prompt

• Paste the following instructions:

You are an AI Agent that must follow these rules:

1. Always use rag_text to retrieve relevant information from the attached document collection before answering.

2. Base your answers on the retrieved content and cite alignment or conflicts clearly.

3. When budgets or constraints are provided, enforce them strictly.

4. When asked for structured output, return valid JSON matching the schema exactly.

Upgrade Your Agent with Smart Capabilities

Copy and paste this prompt:

I need to find appropriate holiday gifts for 3 different people:

1. A colleague who loves tech gadgets (budget $40-70)

2. A senior client who wants thoughtful, professional gifts (budget $80-120)

3. A new team member who enjoys cooking and creative hobbies (budget $25-50)

Please:

- Use my company policy guidelines from our RAG collection

- Follow the budget rules strictly

- For each person, give 3 gift options

Then **[PRESS: Send]**

Upgrade Your Agent with Smart Capabilities

In **Config** tab:

1. Check answer

2. Add a new document on “Company Policies” that contains Gift-Giving Guidelines: https://diabengineering.com.au/wp-content/uploads/2019/02/HSE-MAN-001-001-DIAB-Engineering-Policy-Manual-Rev05-All-Policies.pdf)

Upgrade Your Agent with Smart Capabilities

Copy and paste this prompt:

I previously asked you to find appropriate holiday gifts for 3 different people with specific budgets and constraints. Now I need you to create an interactive HTML Gift Recommendation Dashboard to visualize those recommendations.
ORIGINAL GIFT REQUEST:
I need to find appropriate holiday gifts for 3 different people:
1. A colleague who loves tech gadgets (budget $40-70)
2. A senior client who wants thoughtful, professional gifts (budget $80-120)
3. A new team member who enjoys cooking and creative hobbies (budget $25-50)
Please:
- Use my company policy guidelines from our RAG collection
- Follow the budget rules strictly
- For each person, give 3 gift options
- Include JSON output with:
- gift_name
- price

Then **[PRESS: Send]**

Upgrade Your Agent with Smart Capabilities

Download the html file

Explore, improve

Copy and paste this prompt:

Modify the html so that I can click on the option, one choice per column/ individual, so that when the gift is selected the budget used updates with the price of that product/ gift only, and there is a button in addition, where you can click after selection, to forward in an email where you need to inform the email address and the name of the person, object plus description about Happy Holidays, here is you gift card, thank you !
Open the generated email writing to google account directly, and instead of the original colors use my company branding kit that you can find here:
https://h2o.ai/company/brand-kit/

Then **[PRESS: Send]**

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Certification

https://www.classmarker.com/online-test/start/?quiz=6kk693b055cdcb23

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Thank You!

Contact

Presenter's Name
Presenter's Title
@ youremailhere
in your/linkedin/here